

SakshamAI

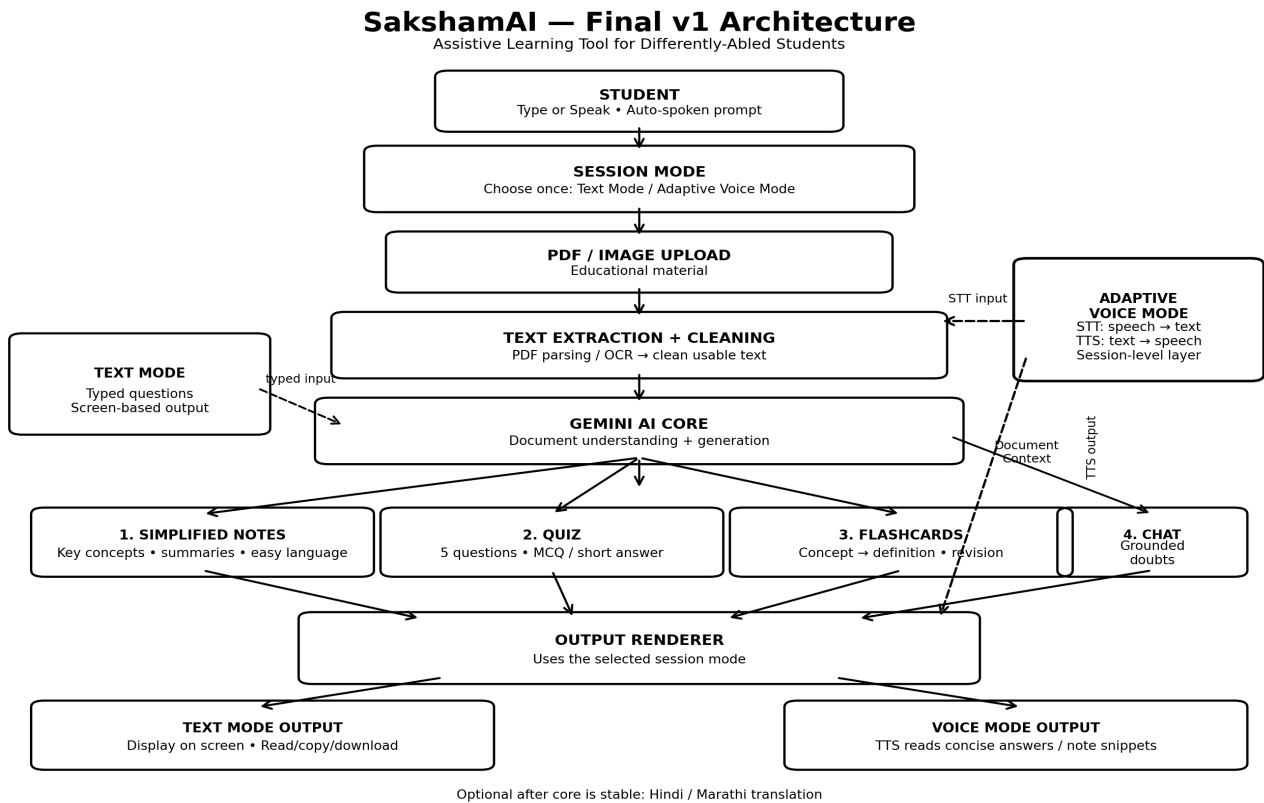
Final v1 — Group Project Plan & Task Allocation

CEP Topic 39: Assistive Learning Tools for Differently-Abled Students

Project goal: Build one reliable AI learning core that converts educational documents into simplified study material and document-grounded doubt solving, with **Adaptive Voice Mode** making the same session accessible through speech.

1. Final Architecture

The architecture is intentionally simple: document processing and Gemini form the core; notes, quiz, flashcards and chatbot reuse the same document context; Adaptive Voice Mode acts as the accessibility layer around the session.



Important: STT and TTS are implementation components of Adaptive Voice Mode, not separate project modules.

2. Final v1 Scope

Priority	Feature	What the team must build
1	Document Processing	PDF/image upload, text extraction/OCR where required, cleaning and preparation of document context.
2	AI Study Support	Gemini generates simplified notes, a 5-question quiz and flashcards from the extracted material.
3	Document-Grounded Chatbot	Student asks doubts and receives answers based on the uploaded document/context.
4	Adaptive Voice Mode	Session-level Text/Speech choice; STT for questions and TTS for concise answers and note snippets.
5	Optional Translation	Hindi/Marathi output only after the core system is stable.

3. What We Will NOT Build in v1

- **Peer-learning network / live student matching** — unnecessary complexity and moderation/privacy issues.
- **Teacher or mentor dashboard** — future institutional feature.
- **Learning profile / personalization engine** — useful later, but not required to prove the core concept.
- **Standalone OCR/STT/TTS module** — these are implementation components, not separate modules.
- **Extra AI features** such as recommendation engines or automatic study plans — only after the core is stable.

4. Proper Task Allocation — 4 Team Members

Each member owns one major area. Members can help one another, but ownership prevents the common problem where everyone works on everything and nobody is responsible for finishing a component.

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Member 1 — Research & Requirements	Member 2 — Gemini AI Core	Member 3 — Chatbot & Voice	Member 4 — Document Processing, UI & Documentation
Main role: Understand the real student problem and convert it into requirements.	Main role: Build and test all Gemini-based learning outputs.	Main role: Build the doubt chatbot and Adaptive Voice Mode.	Main role: Make documents usable by the AI, build the interface and maintain project documentation.
Tasks: • Prepare 8–10 question survey • Conduct student/field interviews • Collect accessibility-related requirements • Analyse responses • Prepare Chapter 4 problem-validation material • Maintain requirement list	Tasks: • Create Gemini integration • Prompt for simplified notes • Prompt for 5-question quiz • Prompt for flashcards • Prompt for concise speech answers • Test output quality and hallucination handling • Maintain prompt files	Tasks: • Implement document-grounded Q&A; • Maintain conversation context • Implement session Text/Speech selection • Integrate STT • Integrate TTS • Add speech-mode fallback/error handling • Test the complete voice flow	Tasks: • Implement PDF/image upload • Implement extraction/OCR and cleaning • Build main UI • Connect UI to team modules • Capture screenshots • Maintain README/report structure • Prepare PPT/demo assets
Final ownership: Research + Requirements + Chapter 4	Final ownership: AI generation + prompt quality	Final ownership: Chatbot + STT/TTS + Voice Mode	Final ownership: Input pipeline + UI + documentation/integration

5. Dependency Between Members

Order	Member	Must provide	Used by
1	Member 1	Requirements, survey findings, target-user needs	All members

Order	Member	Must provide	Used by
2	Member 4	Reliable extracted/cleaned text from PDF/image	Member 2 + Member 3
3	Member 2	Gemini prompts + functions for notes/quiz/flashcards	Member 4 UI + Member 3 chatbot/voice
4	Member 3	Chatbot + STT/TTS + session voice mode	Member 4 integration + final demo
5	Member 4	Integrated interface	Entire team

6. Shared Responsibilities

- **Everyone:** test the complete system, report bugs and understand the full architecture for viva.
- **Everyone:** contribute to report writing, but each member owns their assigned chapter/technical section.
- **Everyone:** rehearse the final demo at least twice.
- **Member 2 + Member 3:** jointly validate that Gemini answers remain grounded in the uploaded document.
- **Member 3 + Member 4:** jointly test the full Speech Mode UI flow.
- **Member 1:** keep evidence of survey/interview work for the report and presentation.

7. Development Sequence

Phase	Work	Owner	Completion condition
1	Requirements + survey + UI sketch	M1 + M4	Scope frozen and sample document selected.
2	PDF/image extraction + cleaning	M4	Reliable text produced from test documents.
3	Gemini notes + quiz + flashcards	M2	All 3 outputs generated from one document.
4	Grounded chatbot	M3	Questions use uploaded context and handle missing information.
5	Adaptive Voice Mode	M3	Speech selection → STT → AI → TTS works.
6	Full UI integration	M4 + All	One complete student session works.
7	Testing + report + PPT + demo	All	Stable presentation build and fallback ready.

8. Final Demo Flow

Step	What happens	Owner during demo
1	System asks aloud: "Would you like Text Mode or Speech Mode?"	M3
2	Student selects Speech Mode.	Presenter / M4
3	Upload prepared PDF.	M4
4	Document is extracted and processed.	M4 + M2
5	Simplified notes are generated.	M2
6	TTS reads a short notes snippet.	M3
7	Student asks one pre-tested spoken doubt.	M3
8	STT → Gemini → document-grounded answer → TTS.	M3 + M2
9	Presenter briefly mentions quiz and flashcards.	Presenter

9. Demo Safety Plan

- **STT failure:** switch to the typed question immediately; explain that the same AI core supports both modes.
- **TTS answer too long:** speech-mode prompt limits answers to about 2–3 sentences.

- **PDF extraction failure:** use a pre-tested PDF/image sample.
- **Internet/API issue:** keep screenshots or a short recording as a last-resort backup.
- **Do not improvise the live question:** use one rehearsed question with an answer present in the uploaded document.

10. Report Chapter Ownership

Section	Primary owner	Support
Problem statement + survey/field findings	M1	All
Literature / existing systems	M1	M2
System architecture + workflow	M4	M3
AI prompts + generation methodology	M2	M3
Chatbot + Voice Mode implementation	M3	M2
UI + document processing	M4	M3
Testing + results	All; compiled by M4	M1 provides survey evidence
PPT + final demo	M4 + Presenter	All

11. Final Team Rule

Freeze v1. Do not keep adding features. The project is successful if the team can reliably demonstrate **document upload → extraction → Gemini study support → grounded doubt solving → Adaptive Voice Mode**. Everything else should remain future scope unless the core is already stable.

12. Final Checklist Before Submission

- Survey and field evidence collected.
- Sample PDF/image documents tested.
- Extraction and cleaning stable.
- Notes, quiz and flashcards generated correctly.
- Chatbot answers grounded in document.
- Text Mode works.
- Speech Mode works.
- Initial prompt can be spoken aloud.
- STT and TTS tested in realistic environment.
- Typed fallback works.
- UI integration complete.
- Screenshots and report evidence collected.
- PPT completed.
- Full demo rehearsed by all members.